

Supplementary Figures

Fig. S1 | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.

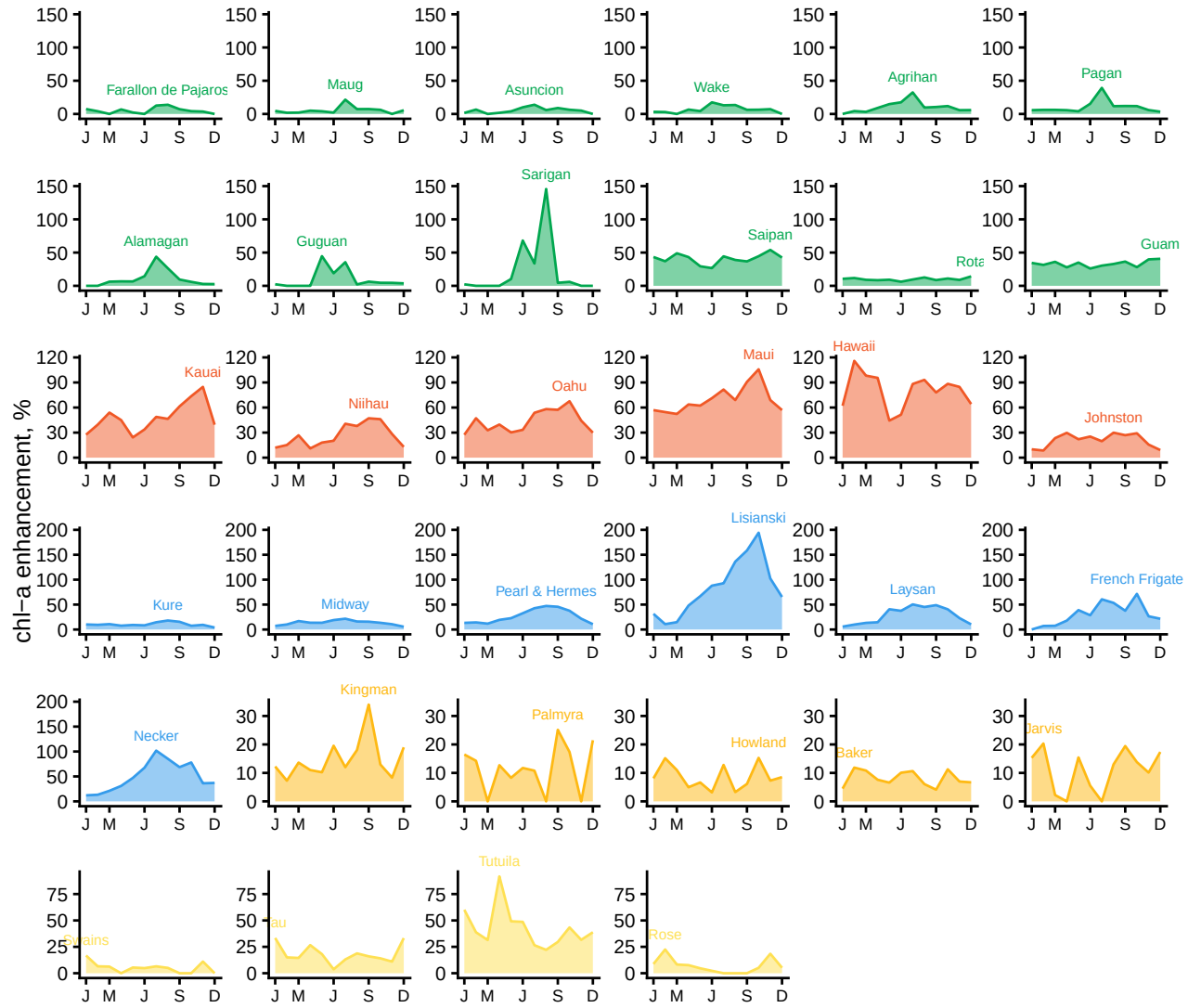


Fig. S2 | Seasonal variation in IME strength at each island. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

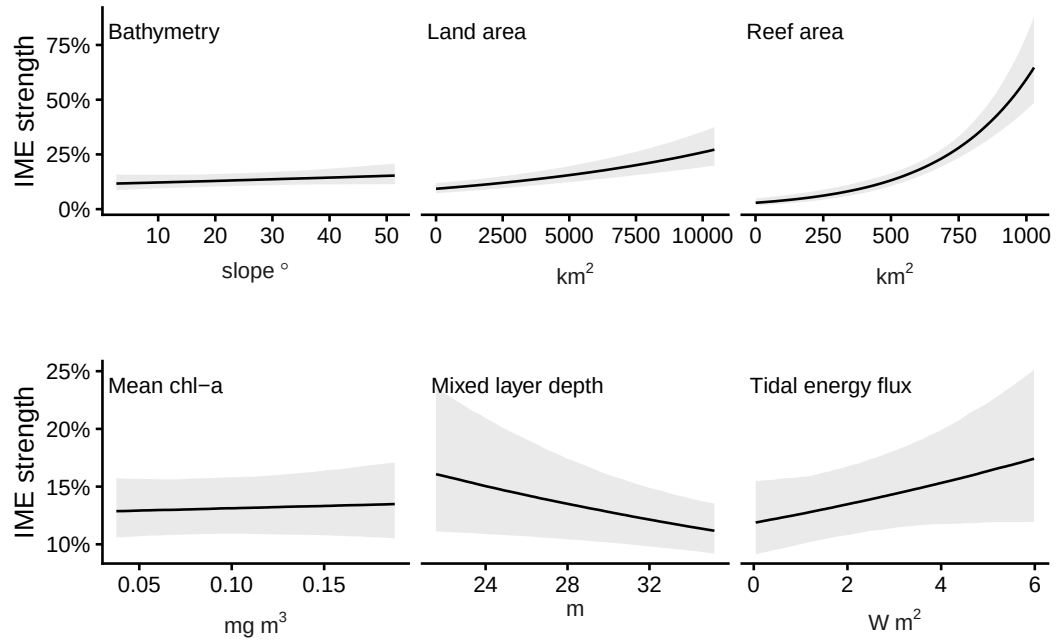


Fig. S3 | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions ($\pm 95\%$ certainty intervals). Note different y-axis values between static and dynamic covariates.

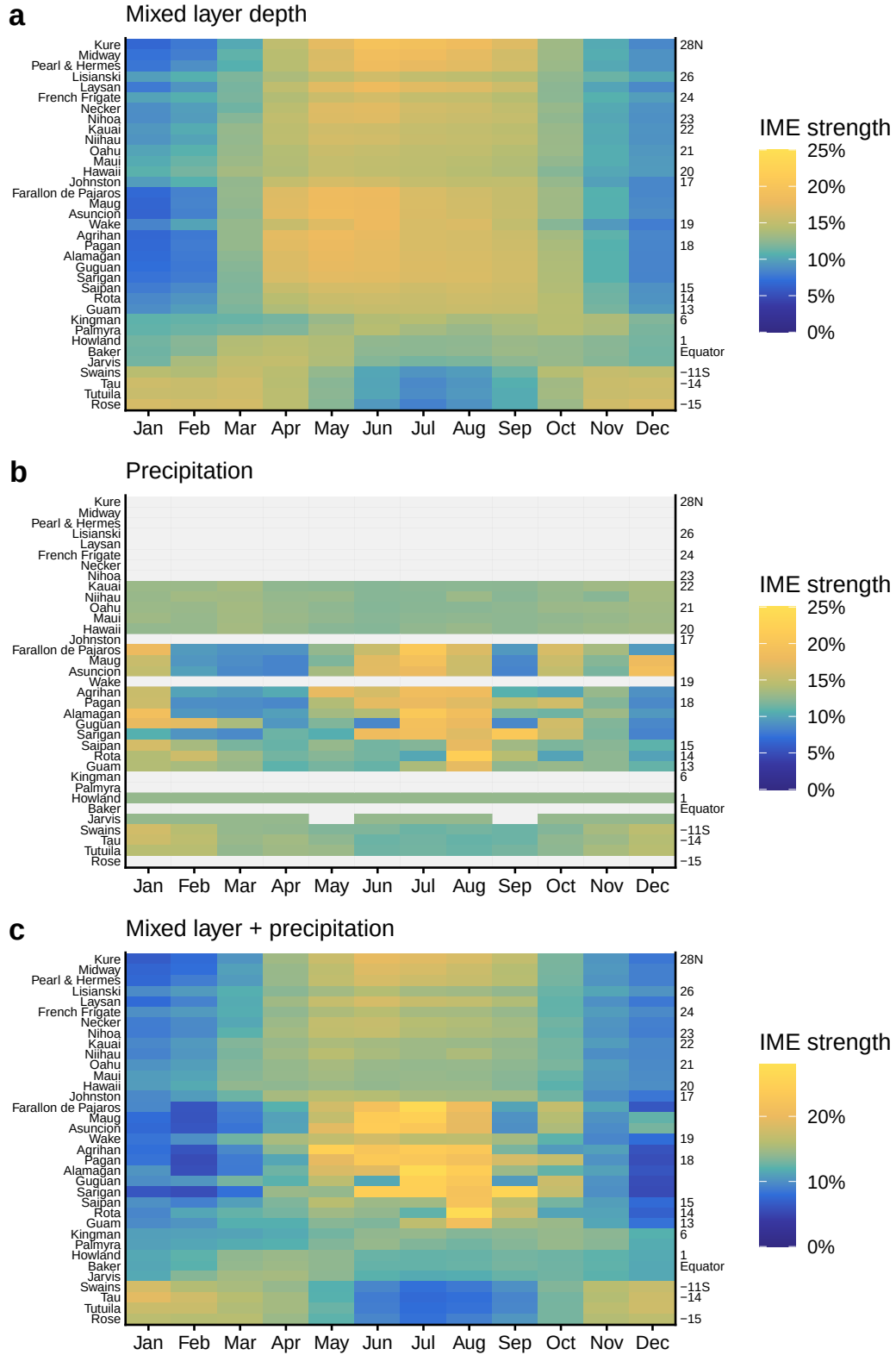


Fig. S4 | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

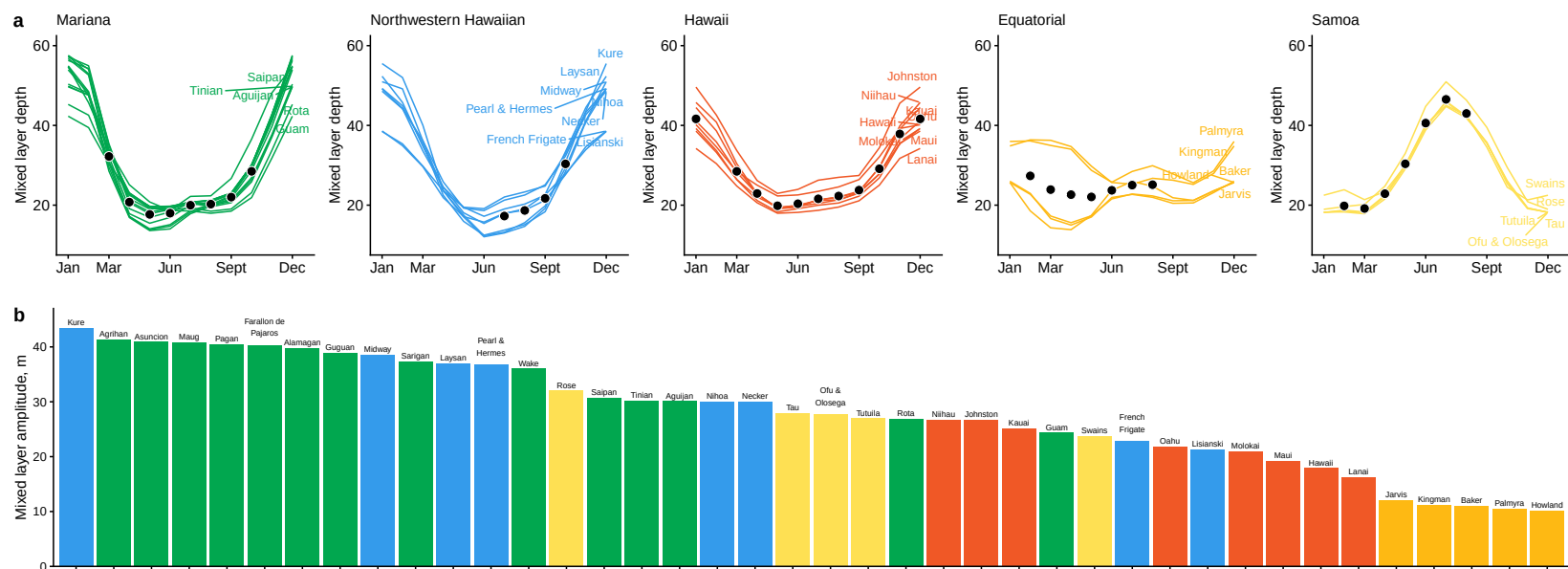


Fig. S5 | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a General Additive Model with smoothers for time and month (cyclic), overlaid with points indicating timing of coral reef fish surveys.

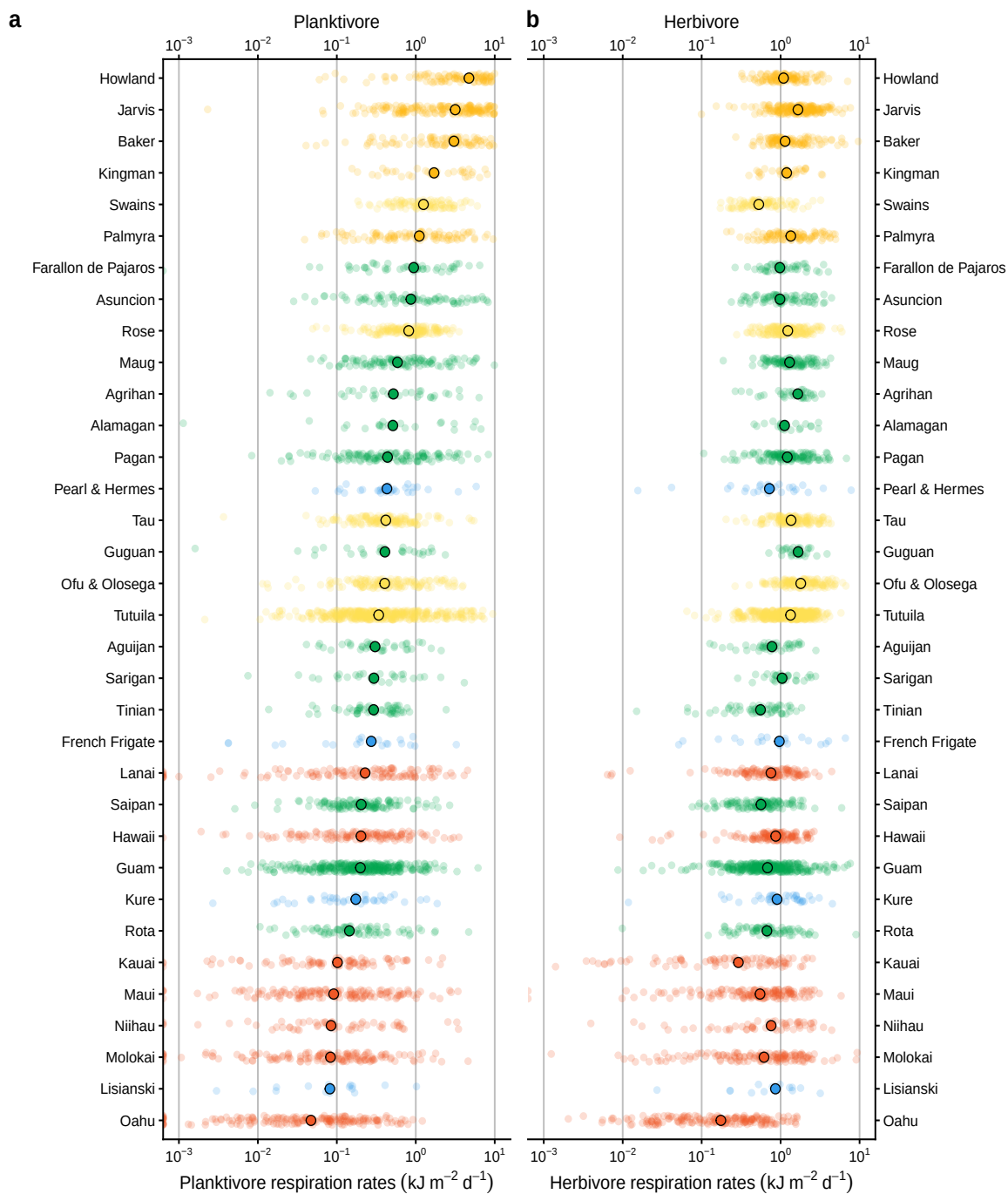


Fig. S6 | Respiration rate of reef-associated planktivores on coral reefs at 34 Pacific islands and atolls. Each point is the median respiration at each island, with jitter showing unique sites ($n = 3,622$).

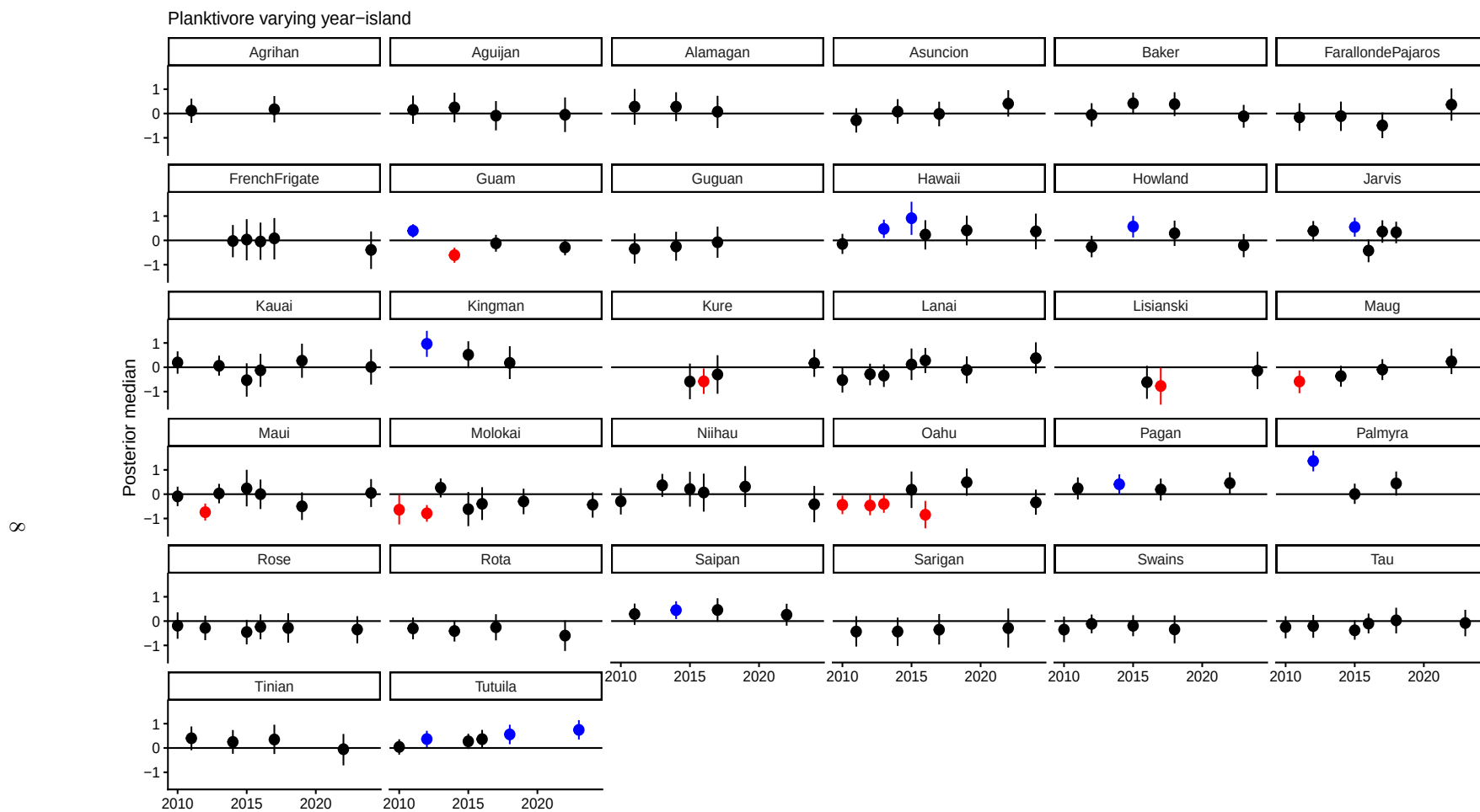


Fig. S7 | Effect of survey year on planktivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

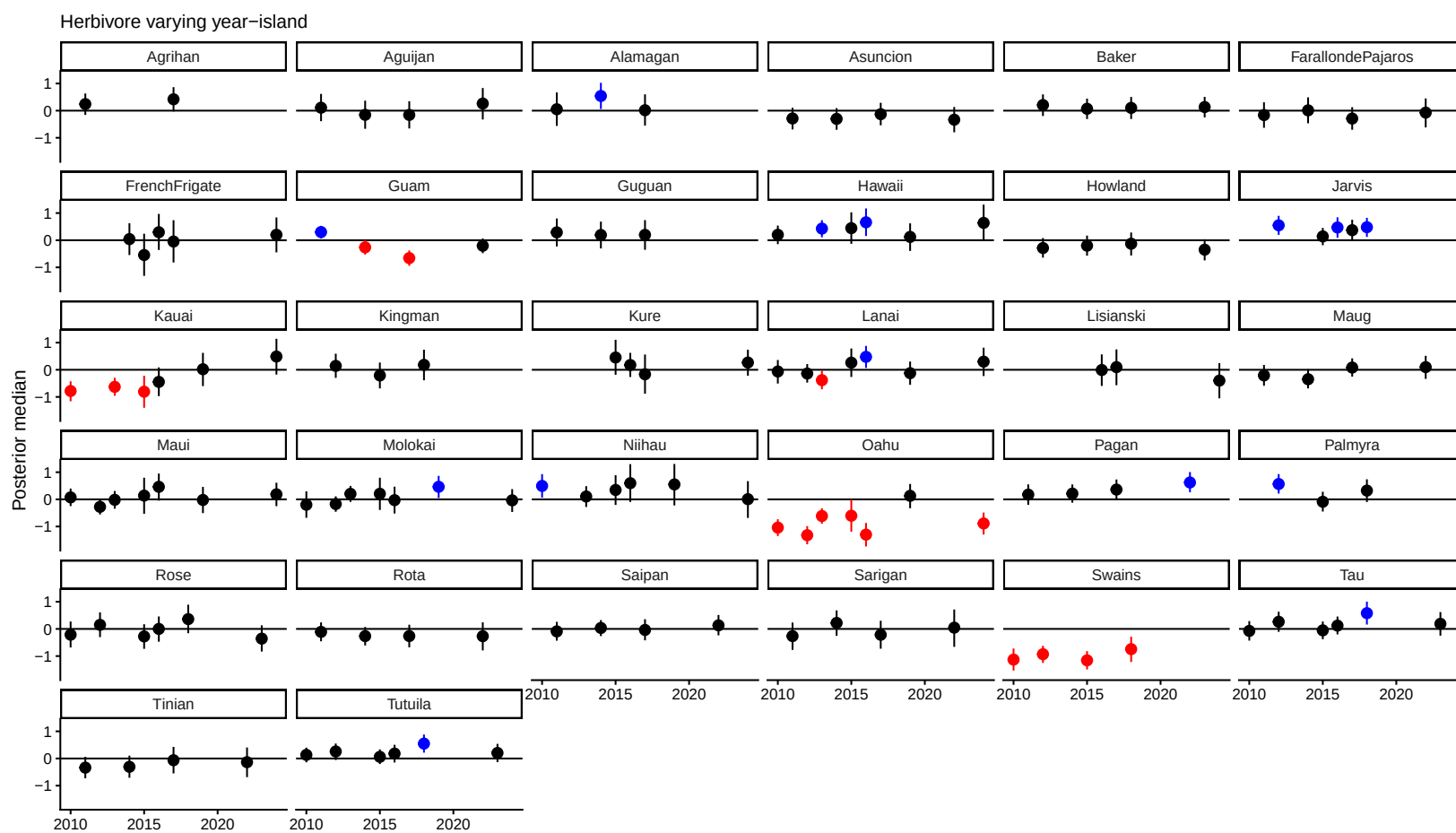


Fig. S8 | Effect of survey year on herbivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

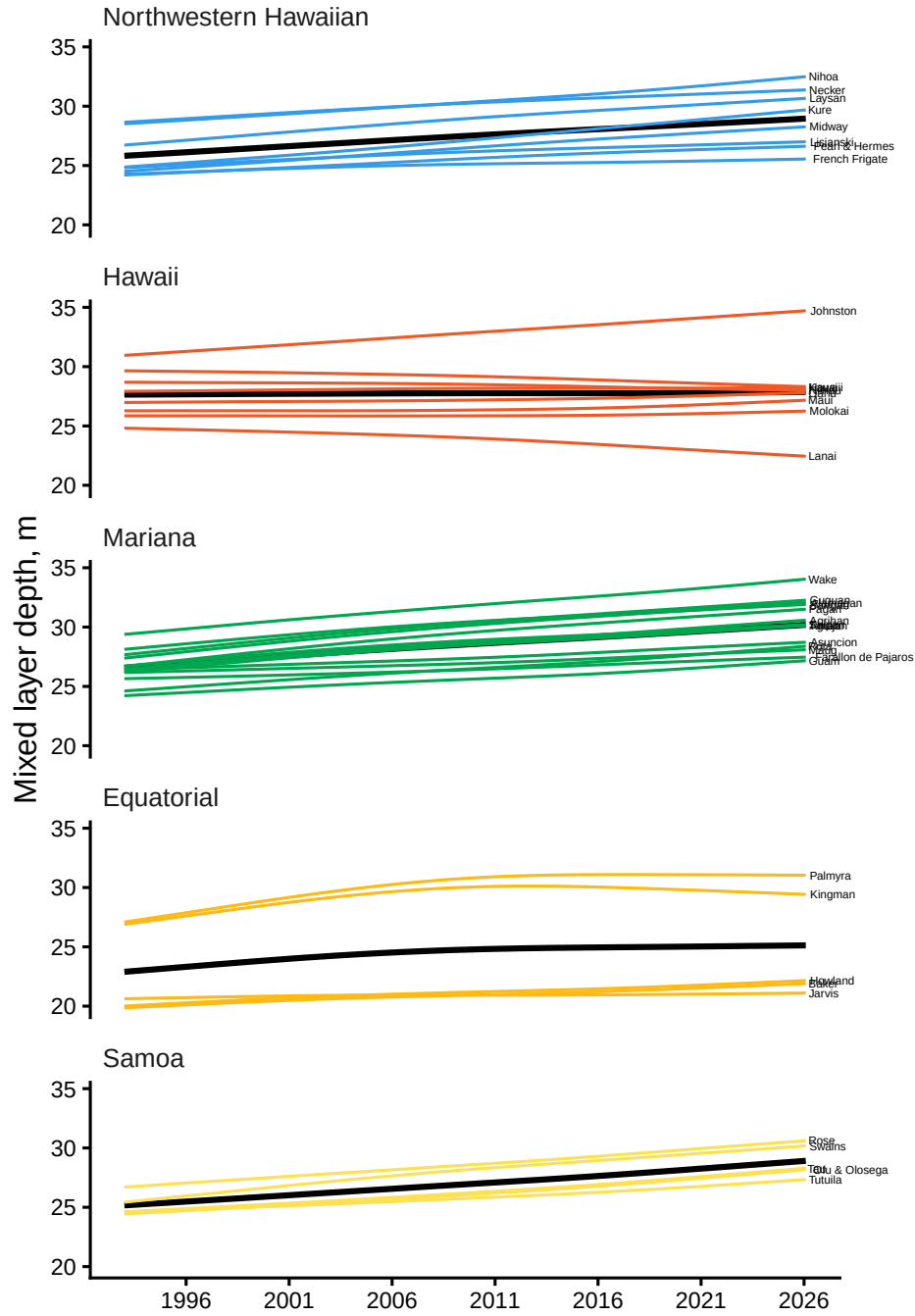


Fig. S9 | Island-level temporal trends in the mixed layer depth layer from 1993-2026. Lines are mean predicted mixed layer, accounting for monthly variability, from a generalised additive model, for each region (black with 95% CI filled by region) and overlaid with mean island predictions (coloured lines).

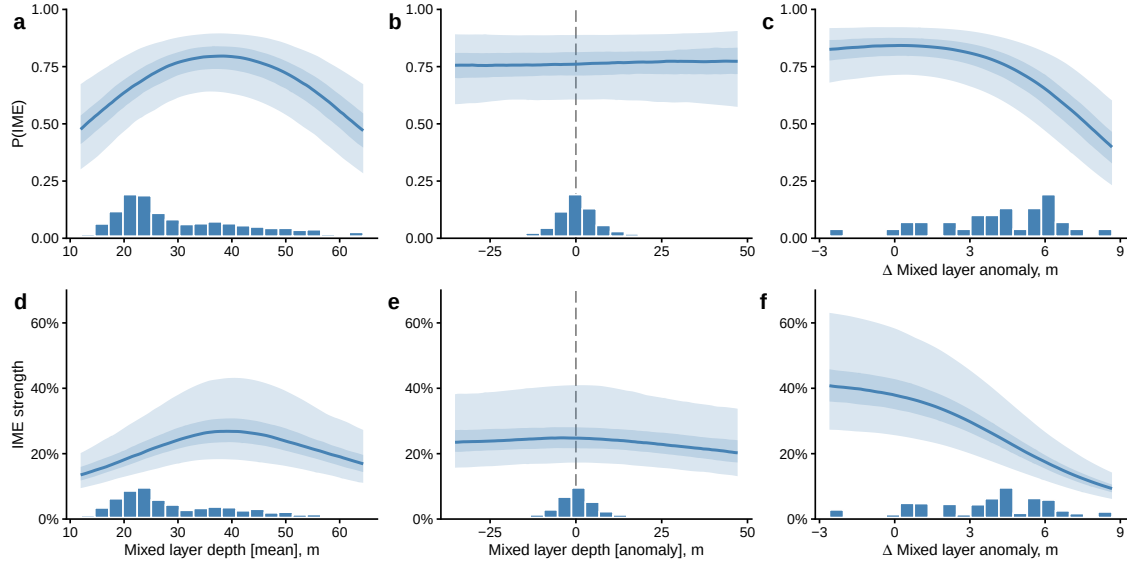


Fig. S10 | Effects of mixed layer depth covariates on IME dynamics from 1997-2025. Top-row is probability of detecting an IME and bottom-row is IME strength (%), for mean mixed layer depth (a, d), mixed layer depth anomaly (b, e), and the mixed layer change from 1993-2026 (c, f). Lines are median posterior estimates with 95% and 50% certainty intervals. Histograms show underlying data distribution of each covariate along x-axis.

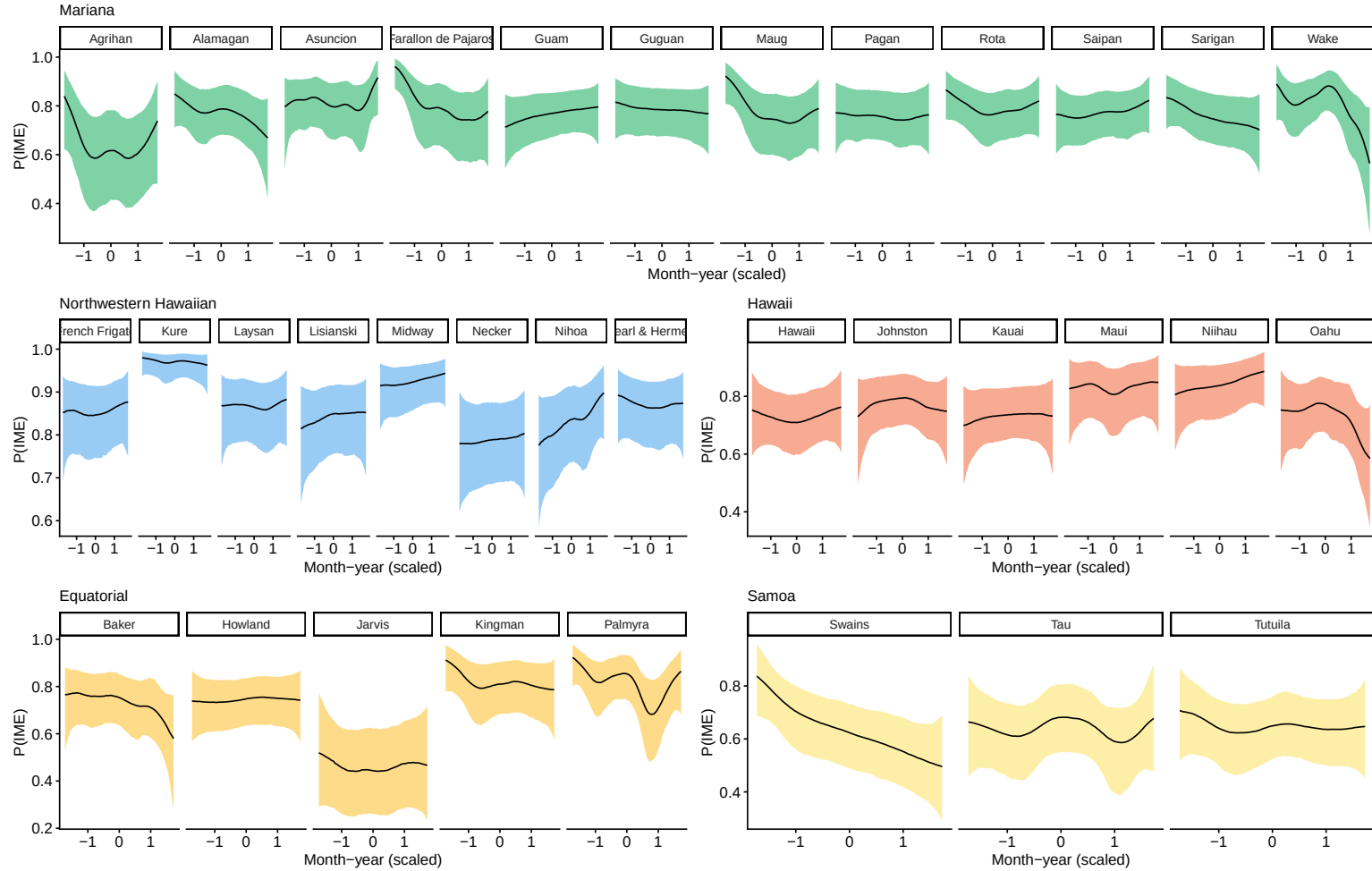


Fig. S11 | Temporal change in the probability of IME occurrence at each island from 1997-2025. Lines are median posterior estimates with 95% and 50% certainty intervals, excluding effects of mixed layer covariates and month smoother.

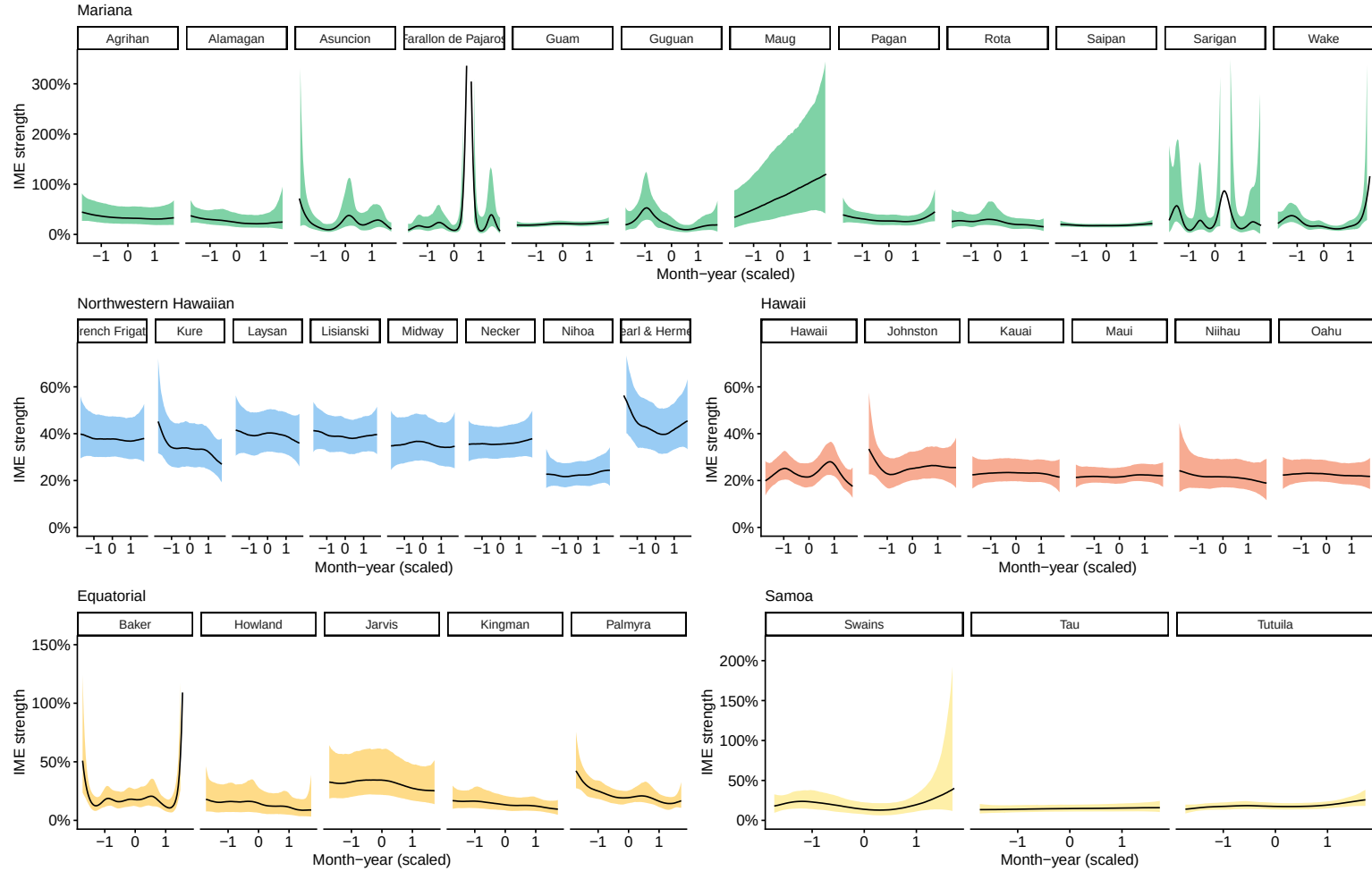


Fig. S12 | Temporal change in the IME strength at each island from 1997-2025. Lines are median posterior estimates with 95% and 50% certainty intervals, excluding effects of mixed layer covariates and month smoother.